

Overview

Released in 2025, the Morpilot HS009 Milk Frother is a compact countertop kitchen appliance designed to deliver professional milk preparation. With six selectable modes, it allows users to prepare café-style drinks like cappuccinos, lattes and hot chocolate. Its durable build, quiet operation and internal markings support both precision and convenience whilst remaining easy to use. The device's variable capacity is desired makes it suitable for solo use or sharing, while its sleek form factor complements modern kitchen environments.

Styling

First Impressions

Minimal Is Modern

The frother has been designed to give a minimalist and modern impression with a sleek, cylindrical form that feels clean and deliberate.

The icon-based control panel uses subtle illuminations to communicate function without clutter. Meanwhile the spout and handle also follow the minimalist design language with their compact geometry and smooth integration into the body.

The handle's shape and placement resembles the silhouette of a mug, touching on the product's domestic context. The way this product has been designed, to have as few external features as possible, makes it feel intuitive, approachable and modern.

The Spout

Small Is Not All

The spout's compact form has been designed to avoid disrupting the frother's clean silhouette, similarly to the lid's low-profile grasp. Despite this, the user can still pour milk out accurately and cleanly.



Spout allows milk to be poured out easily and without spilling.



Plain Packaging

The packaging features little visual detail, with only the frother appearing on the front with a very plain colouring. This subdued aesthetic suggest that this product is primarily sold online, where shelf appeal to potential buyers is less crucial.



Very little colour and a plain aesthetic on the box.

Packaging That Undermines Confidence

When unboxed the frother is contained in a cheap plastic cover. This poor initial interaction between the product and user may lead them to question the device's quality and reduce their confidence in the frother from the start.



Frother and cable came in a cheap plastic bag.

The Lid

Clear View Builds Trust

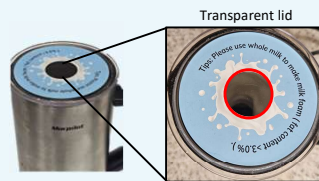
Being able to see the milk as it warms and froths, not only offers visual reassurance all is working correctly but it builds trust and reduces uncertainty, especially for first time users.



Sealing Grips Spills Skip

A silicone sealing ring secures the lid in place, preventing vibration and rattling during operation, observed when the machine was used without the sealing ring.

This feature also helps contain heat and prevent spillage when the milk is frothed at high speed.



Transparent lid

Slim Grip Suits The Style

A low-profile grasp preserves the frother's clean lines and modern look and is a commonly found aesthetic found on a variety of brands, such as the Morpilot's MF-MP01 model or the Salter Chocolatier frother.

However, it does make it more difficult to remove the lid, especially when using a high temperature mode where the frother's body heats up significantly.



Morpilot HS-009



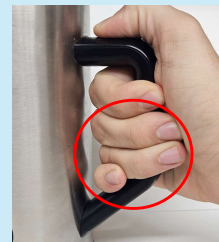
Salter Chocolatier

Morpilot MF-MP01

The Handle

Plastic Stays Cool When Heat Builds

The handle uses a plastic material with a distinct feel from the stainless-steel body. This choice lowers production costs while keeping the grip safe to touch during operation, improving comfort and usability for the user.



Angular Shape Adds Edge

The handle's trapezoid design gives the frother a sharp, modern look and is easy to manufacture. Its compact form blends well with the overall design and prevents slippage.

However, the enclosed shape restricts the use of this device for individuals with larger hands.

Heating Chamber

Clean By Design

The internal heating chamber features a non-stick coating that provides the user with an ergonomic experience. Heated milk and foam slide off the surface smoothly, preventing milk build up and hence reducing the risk of scorching.

On top of this, the coating enables easy maintenance as it resists staining, enabling easier cleaning between uses.

This feature is a subtle and practical design choice that can be found on many modern milk frothers.

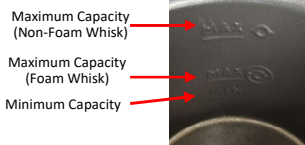


Whole internal heating chamber has a non-stick coating applied.

To See or Not To See Indicator Levels

Level indicators mark the minimum and maximum fill lines for foam and non-foam whisks. They are meant to give the users a quick visual guide and prevent the device from overflowing without the need for manual measurements or guesswork.

However, these markings are difficult to distinguish, a common problem in many milk frothers such as Morpilot's MMF-809 or Salter's Chocolatier frothers.



Maximum Capacity (Non-Foam Whisk)
Maximum Capacity (Foam Whisk)
Minimum Capacity

The Whisks

A Whisk Too Many?

The milk Frother comes with 2 whisks, each intended for a different mode. This is essential as a part of the frother's versatile and multi-role design.

However, when a non-foam mode was used, the frother heated the milk just as effectively without the non-foam whisk as it did with it.



Foam Whisk

Non-Foam Whisk

Effortless Attachment

Both whisks fit onto a stationary shaft within the heating chamber and magnetically connect to the base.

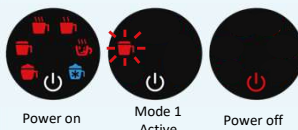
This has been designed to make it easy for the user to remove and wash them, or to swap them out and use different modes.

The Display

Clear Lights Mean Clear Modes

The illuminated display adds to the frother's sleek and minimalist aesthetic whilst offering clear intuitive feedback. It has been designed such that when it is turned off, the display blends into the body and when active, it clearly shows the current mode, power status and operation.

However, some modes have icons with very similar designs, as seen below, which could confuse users, especially those who suffer from poor eyesight.



A Clear Differentiator

The display and metallic, modern design of this frother allows it to stand out in comparison to competitors. Many other products, such as the Salter Chocolatier, use more physical buttons which look less appealing than a digital display.



Frother with Buttons vs Frother with Display

Power Supply



Direct Plug Keeps It Simple At A Cost

This frother has been designed with a direct AC power supply, reducing costs and complexity.

Fewer components make it more compact than other devices such as the Morpilot MMF-809 or Salter Chocolatier.

However, this fixed cable has not been designed ergonomically as it restricts the user's movement when attempting to pour milk out.

Smart Features

Systems for Safer Sips

The frother has a Strix in-built temperature control system for improved reliability, automatically shutting off when the milk or foam is within a pre-set temperature range. This feature prevents burning and over-frothing, enabling the user to enjoy a safe and worry-free experience when using this device. This results in the milk having a non constant preparation time.

Sound on, Milk Done

The frother has a built-in sound system that rings when the device has finished operating. This ergonomic feature allows users to be notified audibly when the milk is ready, eliminating the need for constant user monitoring.

6 Modes in 1

Different Modes for Different Desires

The frother has been designed to offer the user a wide range of different modes. Each provides varying levels of foam and temperature, on top of this there is a dedicated mode to prepare hot chocolate.



Alternative Process Required For Hot Chocolate

This product has a dedicated hot chocolate mode as this process requires the milk and solid chunks of chocolate to be heated prior to whisking to ensure that the powder melts correctly.

User Journey

Prepare all you need to use the frother.



Easily places the whisk inside chamber thanks to magnetic contact.



User fills frother with milk but struggles to fill to the right amount due to difficulties seeing the limitation marks.



User places lid on the frother, needs to apply force to fit it in.



User easily selects mode that they want to use.



Milk gets heated and frothed, user must wait until the bell rings signalling its finished.



User has difficulty removing the lid due to its small profile and the high temperature of the frother's body.



8

User pours out milk and foam easily into a cup.



Market Analysis

A Dodgy Decoy?

The HS009 frother may be a dodgy decoy within the Morpilot range. A small difference in price can be seen between alternative products by the brand and this model despite it missing key features such as a charging platform. This may subtly encourage users to spend a bit more for models that are more practical to use.



HS009	MMF-809	HS-001	MF-MP01	HS010-S
£29.99	£34.99	£35.99	£39.99	£46.00

Competitor Analysis – Budget Frother, Budget Experience

To assess how the Morpilot HS009 compared to rival products in the market performance wise, 6 users took part in a trial where they used 4 different frothers of varying prices. Hot chocolate was made in each device and each received a score for the heat, the time taken, the taste and how easy it was to use (ergonomics) which all then averaged to create a User Experience Score. The results of this procedure can be seen below with the Morpilot HS009 ranking lowest overall.

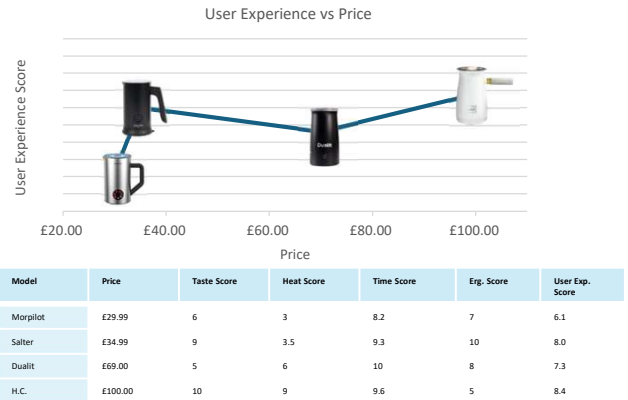
Products Tested

Morpilot HS009
RRP: £29.99
Capacity: 150ml to 350 ml
Variety: 6 Modes

Salter Chocolatier
RRP: £34.99
Capacity: 150ml to 240 ml
Variety: 4 Modes

Dualit Milk Frother
RRP: £69.00
Capacity: 350 ml
Variety: 1 Mode

Hotel Chocolat (H.C) The Velvetiser
RRP: £100.00
Capacity: 150 ml
Variety: 1 Mode



Improvements

Investigating The Lid's Stiffness

Removing the lid from the frother requires excessive force when compared to other frothers such as the Salter Chocolatier or Morpilot MMF-809.

Initially, I attributed this to the lid's compact profile, which offers limited surface area for gripping.

To test this, I used Inventor CAD Software to model a new grip inspired by the Morpilot MMF-809's lid then 3D printed the part and fitted it to the frother.

Despite the improved grip, removal remained difficult. This led me to conclude that the sealing ring itself was the cause of this problem.

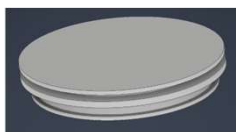
No Need To Grip The Hot Body

This improvement may not have solved the initial problem however the larger profile enables the user to be able to remove it without the risk of touching the hot body of the frother, hence reducing the burn risk posed by it.

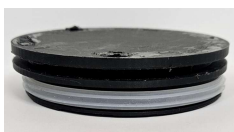


Morpilot MMF-809 Lid.

Multi Layered top section of lid improves user grip



Custom lid modelled and designed in Inventor CAD.



New lid printed out with sealing ring fitted.



Current Lid: hand in contact with hot metal body.



Custom Lid: hand in contact with lid but not the hot metal body.

Designed to be removed

The frother's lid features a sticker which clashes with the design, is hard to remove and leaves mark. The advice on it may be useful for first time users but a lot less so for the more experienced.

Having a low tack vinyl sticker on the lid instead could be an improvement to the frother as it sticks well to metal and plastic however it can be removed easily.

Furthermore, being able to remove this label would give the user the feeling of no longer being a beginner in using the frother.



Storage Whisk?

The frother has two whisks included however there is no way to store them when they one is not in use. A clip built into the lid to store the whisk not in use could be beneficial in this respect to ensure that parts are not lost.

This feature combined with a more removable sticker on the lid would allow the whisk to be stored whilst still enabling the user to see the milk being frothed/heated within.

The identical bases of the whisks would mean one clip would work for both



More Visible is More Usable

The level indicators inside the heating chamber are difficult to see, which makes it hard for users to fill the frother to the correct amounts.

Extending the protrusion to follow the full diameter of the chamber would improve visibility and help users identify the fill limits more easily.

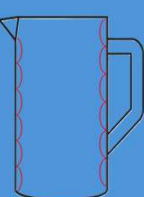


Level Indicator Protrusions go around the whole diameter of the internal heating chamber.

Extra Layers To Keep The Heat In

The body of the frother heats up during use, especially in longer modes such as hot chocolate. This poses a burn risk to the user and makes the device ergonomically difficult to handle when hot.

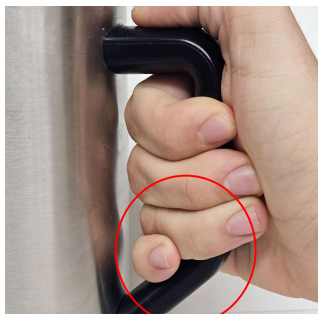
Tastefully adding a layer of insulation to the exterior could reduce the risk this poses to the user whilst not impacting the minimalist design of the frother. This improvement would need to account for the digital display as well as the power supply port when designing it.



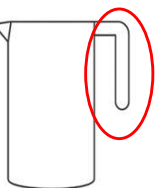
Insulation along exterior

Hold For All Hands

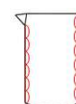
In its current form, the handle is difficult to hold for people with larger hands as the sloped section interferes with the natural positioning of the user's fingers.



A solution to this to make the design both more inclusive and ergonomic would be to replace the handle with a different shape. An open handle would allow more users to be able to use it more easily.



Getting rid of the handle completely and adding a layer of insulation with grooves for the user to hold on could also be a solution. This is a common design used by many other brands such as the Lavazza A Modo Mio milk frother (seen below).



Insulation along exterior with grooves for improved user grip.



Operations

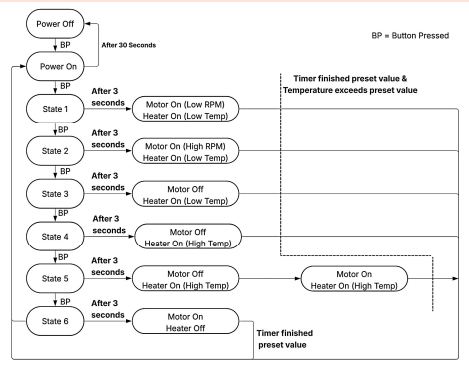
Finite State Machine Architecture

The milk frother offers six different operating modes by adjusting motor speed and heater output to achieve different preparation styles to accommodate the variable needs of a user.

A Finite State Machine architecture has been used to implement this: the user selects a mode via the side mounted interface.

A signal is sent to a Motor and Heater control PCB, instructing it to transition into a specific state, which allows it to then output instructions defining the behaviours of the motor and the heater.

This approach keeps the system versatile while maintaining a simple, reliable control structure.



State flow diagram for the milk frother's operations.

State/Mode	Whisk Motor	Heater
1. Foam Milk (Dense)	On (Low RPM)	On (Low Temperature)
2. Foam Milk (Airy)	On (High RPM)	On (Low Temperature)
3. Hot Milk (High Temperature)	Off	On (High Temperature)
4. Hot Milk (Low Temperature)	Off	On (Low Temperature)
5. Hot Chocolate	On (High RPM)	On (High Temperature)
6. Foam Milk (Cold)	On (High RPM)	Off

State Table showing the behaviours observed by the whisk motor and the heater based off the state the frother is in.

Components

Interior Layout and Overview of Subsystems

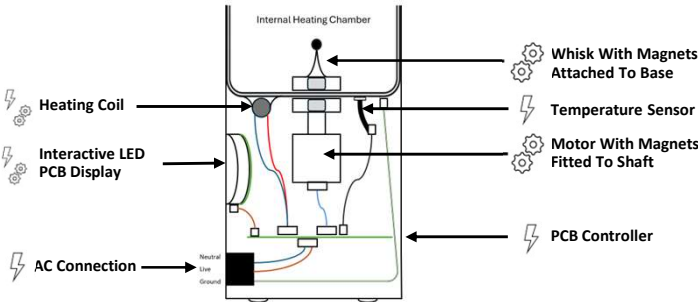
Overview of Section

This section explores key subsystems which are essential to the functionality of the frother, enabling it to heat up milk whilst also frothing it to give it a quantity of foam desired by the user.

Mechanical and Electrical Components

The frother system located beneath the internal heating chamber is comprised of a variety of different components, both mechanical and electrical in nature.

These different parts work together to deliver precise heating, consistent frothing and intuitive user control.



Component Separation

The design of the frother separates the lower section that contains the motor, PCB and other electronics from the internal heating chamber to prevent heat transfer and liquid ingress.

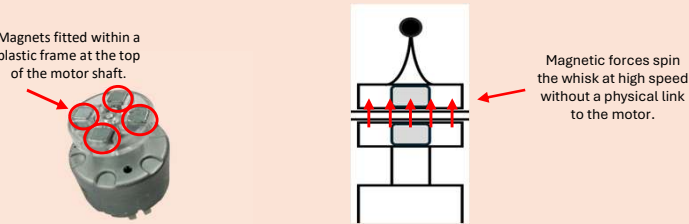
This split layout protects sensitive components from thermal and moisture damage and supports safe, reliable operation over time.

Magnetic Contact Motor

Magnetic Contact Prevents Contamination

A motor beneath the sealed heating chamber spins magnets that engage with those built into the whisk, rotating it at high speed.

In addition to its ease of removing or replacing the whisk, this feature plays a crucial functional role. This sealed design protects internal hardware from liquid ingress.



A High-End Feature On A Low-Priced Product

Magnetic coupling systems are widely used in milk frothers, found in a large number of competitor products.

However, this feature is most commonly found in higher-priced models such as the Nespresso Aeroccino 3 or the Lavazza A Modo Mio Frother which can have prices which exceed that of Morpilot's by a factor of 2 or 3.

Nespresso Aeroccino 3 Milk Frother
RRP: £91.00



Extrusion which holds the whisk in place when rotating. This is similar to the one used by the Morpilot frother.

Lavazza A Modo Mio Milk Frother
RRP: £69.00

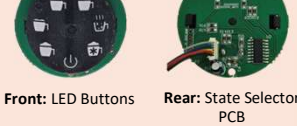


Due to the diameter of the whisk being the same as that of the internal heating chamber. No extrusion is needed to fix the whisk in place.

Interactive LED Interface

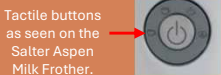
State Selector and Emitter

The display consists of seven LED touch-buttons which interface with a small PCB mounted behind the display, which interprets the user's input and sends the corresponding state signal to the Motor and Heater control PCB.



Front: LED Buttons
Rear: State Selector PCB

Improvement



Tactile buttons as seen on the Salter Aspen Milk Frother.

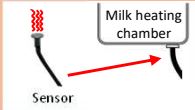
Tactile Rather Than LED Buttons

Replacing the LED touch-buttons with tactile mechanical buttons would simplify the user interface by allowing each mode to be selected directly, rather than cycling through a single illuminated button.

Mechanical buttons would also reduce reliance on display-specific electronics, lowering component complexity and freeing cost that could be redirected toward other functional areas of the product.

Temperature Sensor

Monitoring Milk Temperature



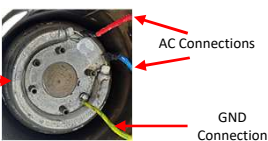
A heat sensor is fixed to the underside of the internal heating chamber to determine the temperature of the milk being heated.

However, positioning the sensor on the outside of the heating chamber means it measures the wall temperature rather than the milk. This indirect reading is less accurate due to external factors like ambient temperature, airflow and thermal lag.

Heating Coil

Metal Heats Milk

An electric heating coil is built into the bottom of the internal heating chamber. This means that when the frother is turned on, the internal heating chamber conducts the thermal energy provided by this coil straight into the milk. The Voltage output by the PCB will dictate the temperature to which the coil will be heated.



Electrical Safety Feature

Since the coil is mounted directly onto a conductive metal base, a separate Ground (GND) connection is essential to ensure user safety by preventing the metal surface from becoming live in case of a fault.

However, this results in a high level of heat being emitted through the sides of the milk frother and wasted, hence making the frother very hot to the touch when held by the user.

Improvement

Improved Insulation for Improved efficiency

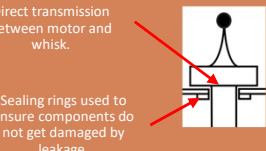
Applying an insulating layer to the exterior of the frother would enhance user safety while improving overall thermal efficiency.

By limiting heat loss through the housing, more of the coil's energy is retained within the milk, allowing it to reach the target temperature more quickly and with reduced energy consumption.



Thicker insulation would allow less heat to leave the frother, providing improved efficiency and reduced usage times.

Improvement



Direct transmission between motor and whisk.

Sealing rings used to ensure components do not get damaged by leakage.

Direct Transmission To Lower Costs

A direct transmission shaft, used in models like the Salter Chocolatier, simplifies the mechanism and lowers production costs. A sealing ring prevents leakage into electrical components, though this design sacrifices the full isolation offered by magnetic coupling.

Adopting this more conventional mechanism could reduce complexity and enable cost savings, allowing other aspects of the design to be improved.

Whisk Head Deep Dive

A Brief Introduction To Foam

In order to understand how whisk geometry influences the quality of milk foam, a deep dive was undertaken to understand how different designs affect the way that air and milk are mixed. Foam forms when air is pulled into the liquid and broken into bubbles, which can create either smooth microfoam or lighter macrofoam. Key qualities such as bubble size, density, texture, sheen and stability depend on how effectively the whisk aerates and circulates the milk. By testing each whisk under consistent conditions, this deep dive aims to identify which designs produce the most desirable foam.

Different Whisks With Different Design Considerations

Adapting to Existing Conditions and Testing

To investigate how whisk geometry affects foam quality, custom heads were designed and fitted to the frother base for direct comparison under identical conditions. The original whisk was modified by cutting off its stalk so each prototype could be mounted securely, allowing all designs to be tested using the same mechanical setup.

To conduct this investigation, each whisk was tested under the same conditions by using the same milk preparation settings to investigate the impact different shapes or features would have upon the milk whisking.

The Original Whisk

The original whisk uses a simple coil mounted on a plastic base with a central stalk, a widely used design found on frothers at every price point.

Stalk improves circulation, creating a stronger vortex for improved mixing.



Radial Blade Whisk	Winged Flow Whisk	Angled Winged Flow Whisk	Arrowhead Whisk	Conical Whisks
Design Considerations This whisk features a circular hub with multiple flat blades arranged radially. The high blade count may increase shear and surface contact, promoting strong outward mixing and dense, uniform agitation across the base of the frother.	Design Considerations This design uses two flat, wing-like extensions mounted on a central hub. The wings may scoop and redirect milk as they rotate, which may strengthen the vortex and improve overall circulation for smoother, more consistent mixing.	Design Considerations Mounted at an angle relative to the shaft, this whisk uses its sloped blades to push milk upward as it rotates. The angled geometry may introduce vertical flow components that increase aeration and create more varied circulation patterns compared to flat-wing designs.	Design Considerations This design combines a hollow triangular frame with a central mounting ring, creating a rigid, open structure. The geometry may guide flow through the triangular opening while keeping the whisk steady during rotation, helping to shape the vortex and reduce wobble at the base.	Solid Cone Whisk A smooth conical form with a rounded top that maintains clean rotation but offers limited vortex stabilisation. Perforated Cone Whisk A similar cone with added holes that allow fluid to pass through, increasing vortex stability and reducing wobble during mixing.

Whisk		Process Dynamics			Foam Outcomes				Foam Stability			Whisking Profile
		Time Until Stability	Flow Description	Splashing	Type of Foam Produced	Bubble Size	Texture	Thickness	One Minute (mm)	Two Minutes (mm)	Milk/Foam separation	
Original Whisk		25 Seconds	Stable and uniform, very little turbulent flow	None	Microfoam	Initially small, large after 45 seconds	Thick	High thickness	10 mm	5 mm	No	
Conical Whisk With Smooth Surfaces		1.02 Minutes	Very little turbulent flow once stabilised.	Large amount initially and periodically until stability was reached	Macrofoam	Small	Bubbly	Low thickness	4 mm	2 mm	Yes	
Conical Whisk With Perforations		15 Seconds	Stable but slow, very little turbulent flow	Small amount at start until vortex stabilized.	Microfoam	Smaller than above	Bubbly	Low thickness	5 mm	3 mm	Yes	
Arrowhead Whisk		Did not achieve stability	Low level of turbulence throughout	Small amounts throughout but not excessive	Microfoam	High number of small	Bubbly	Low thickness	3 mm	0 mm	Yes	
Winged Flow Whisk		Did not achieve stability	Medium level of turbulence throughout	Large amounts as whisking progressed	Macrofoam	Large	Bubbly	Low thickness	3 mm	1 mm	Yes	
Angled Winged Flow Whisk		Did not achieve stability	High level of turbulence throughout	Very large amounts	Macrofoam	Large	Thick but unstable	Medium thickness	6 mm	1 mm	Yes	
Radial Blade Whisk		32 Seconds	Once stabilized, there was little turbulent flow.	Small amounts	Microfoam	Small	Foamy	Medium thickness	7 mm	2 mm	Yes	

No Magic Whisk To Meet Every Need

Following this deep dive process into whisks, it is clear that there is no one form that can meet all the different criteria. Foam creation conditions vary greatly based off vortex formation, turbulence and air incorporation as these induce different levels of foam textures, bubble sizes and stability profiles.

- Microfoam is best achieved with designs that stabilize quickly and minimize turbulence, such as the original and conical whisks. This is best suited for drinks such as Lattes or Macchiatos.
- Macrofoam is best achieved from designs generating high levels of turbulence such as the winged flow designs. Drinks such as hot chocolates or cappuccinos are best suited for this type of foam.
- Milk and foam separation is undesirable as it generates an inconsistent texture and ruins the drink's consistency.

Overall, these results show that foam texture and quality depend on a wide range of variables, but ultimately, they must align with the user's desired outcome.

Improvement

Whisks and Modes for Different Foams

The milk frother already has the option to provide 2 different types of foam, "dense" and "airy". By reinvesting costs saved within other design improvements, Morpilot could expand this capability by introducing additional whisk attachments alongside a broader set of programmable modes.

Pairing specific whisk shapes with specific mixing profiles would allow users to create a wide range of foam textures, granting them the ability to match the different requirements of professionally made drinks.

The users would hence be granted greater control over a more versatile and adaptable frother.

- | Whisk attachments | Frothing mode |
|--|---|
| <ul style="list-style-type: none">- Microfoam whisk- Macrofoam whisk- Cold foam whisk- Thick foam whisk | <ul style="list-style-type: none">- Latte Art Mode- Cappuccino Mode- Hot Chocolate Mode- Smooth mode |

Combination

Microfoam Whisk + latte Art Mode = Latte
Microfoam Whisk + Smooth Mode = Flat White
Macrofoam whisk + Cappuccino Mode = Cappuccino
Thick Foam Whisk + Hot Choc. Mode = Hot Chocolate

Overview

The Morpilot HS009 Milk Frother contains 45 components identified through the disassembly of the product. The total manufacturing cost is estimated at £6.63 including the power cable and packaging. Most parts are produced through automated processes while final assembly is completed by hand.

This reflects a standard approach for the manufacture of low-cost consumer electronics and provides valuable insight into how the product's construction is linked to its retail price.

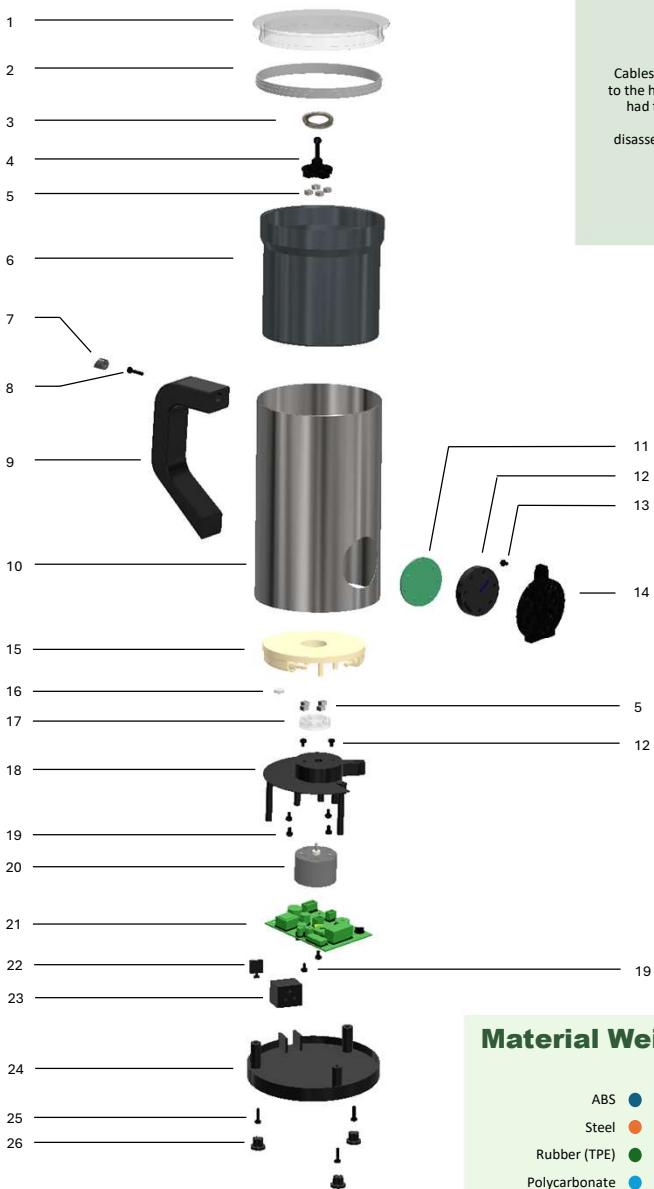
Bill of Materials

Item Number	Section	Part Name	QTY	Material/Sub Parts	Mass (g)	Cost
1	Lid	Lid	1	Polycarbonate	73	£0.16
2	Lid	Lid sealing	1	Silicone Rubber	4	£0.01
3	Whisk	Whisk Coil	1	Stainless Steel Spring Wire	15	£0.02
4	Whisk	Whisk	2	Polypropylene	8	£0.01
5	Whisk	Motor	8	Neodymium	1.5	£0.83
6	Body	Inner Shell	1	Stainless steel	250	£0.30
7	Handle	Handle Screw Cover	1	TPE	3	£0.02
8	Handle	M3 x 14 mm Phillips pan-head machine	1	BOP		£0.01
9	Handle	Handle	1	ABS	180	£0.13
10	Body	Outer Shell	1	Stainless steel	147	£0.17
11	Display	Display PCB	1	Epoxy laminate, Resistors, LEDs, Capacitor, Button	20	£0.50
12	Display	Button Interface	1	ABS, Sticker	15	£0.02
13	Display	Motor	3	M2.5 x 4 mm Phillips pan-head machine screw		£0.03
14	Display	Display Cover	1	Smoked Acrylic	3	£0.01
15	Body	Heating Plate	1	Ceramic, heating coil	200	£0.90
16	Electronics	Strix Temperature Sensor	1	BOP	15	£0.20
17	Motor	Motor Magnet Frame	1	Clear Acrylic	12	£0.12
18	Frame	Inner Component Frame	1	ABS	113	£0.11
19	Frame	M2.5 x 5 mm Phillips pan-head self tapping screw	6	BOP		£0.04
20	Motor	DC Motor (RF-500TB-12560)	1	BOP	200	£0.64
21	Electronics	Main PCB	1	Transformer, Relay, Capacitors, Resistors, Diodes, ICs, Wires	150	£1.00
22	Frame	Socket Cover	1	TPE	8	£0.05
23	Electronics	C5 Cloverleaf Connector	1	BOP	50	£0.20
24	Frame	Bottom casing	1	ABS	187	£0.19
25	Frame	ST2.5 x 10 mm Phillips pan-head self-tapping screw	3	BOP		£0.03
26	Frame	Screw Covers	3	TPE	2	£0.03

BOP: Bought Out Part

Exploded View

Exploded view of the HS009 Frother showing all the components and how they fit together.



Replace Not Repair

The frother was not designed to be taken apart or repaired. This is evidenced by several parts being soldered directly to the inner shell, making removal difficult without damage.

This reflects a design approach focused on quick assembly and low cost, where the product is intended to be replaced rather than serviced.



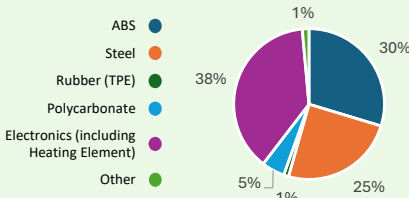
Cables connected to the heating unit had to be cut in order to disassemble fully.

High End Component Usage (16)

The product description lists this part as being provided by **Strix**, a market leader in the *design, manufacture and supply of kettle safety controls* which is applied in this case to a milk frother.



Material Weight Distribution



Cost Analysis

Total Material Cost

Having investigated the component cost of the manufacture of the frother, the value was estimated at **£5.73**. When considering the additional costs of the power supply cable and packaging, this rises to approximately **£6.63** based off the split below:

- Frother - £5.73
- Power Supply cable – £0.80
- Packaging and instructions – £0.10

This total represents 22 percent of the retail price which remains a sensible estimate as small consumer electronics typically allocate between 25 percent and 33 percent of their retail price to manufacturing, often lower for small retailers.

Overall Cost Distribution

Overall Profit - £3.87, 13% of RRP

According to Amazon's 2026 profit margin report, most third-party sellers operate with net margins in the 15–20% range. Morpilot's estimated profit sits close to this band, indicating that this approximation is realistic

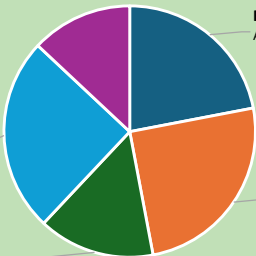
Labour and Production Costs - £7.50, 25% of RRP

The production of this device takes place in China where lower costs are available due to large scale automation and inexpensive labour.

Overhead Costs - £4.50, 15% of RRP

Overhead costs represents logistics, storage and testing. Modern global supply chains are highly optimised for delivering mass produced items, helping keep these prices competitively low.

By considering the product's **RRP price of £30**, Morpilot's limited brand presence and the cost structure of small appliances sold through Amazon, a reasonable estimate of the profit margin for this product can be made.



Raw Material Costs: £6.63, 22% of RRP
Approximation obtained through BoM

Platform (Amazon) Fees - £7.50, 25% of RRP
Includes referral fee and fulfilment charges applied to Amazon retailers, which significantly influences pricing for third-party sellers.

Deep Dive: Body Assembly

1. Deep Drawing of Parts

Outer Shell

The frother's outer body begins as a stainless-steel sheet blanket. This *blank* is then deep drawn into a seamless cylindrical shell, producing a rigid or uniform wall. The rim is then trimmed to its final height and the exterior is brushed to give the product its shiny appearance.

Inner Shell

A second stainless-steel blank is deep drawn to form the inner heating chamber. Only the upper section matches the outer shell's diameter for rim joining, while the lower section is drawn narrower to create the required annular air gap for insulation and internal clearance. Level markers on the inside of the inner shell are also made during this process.

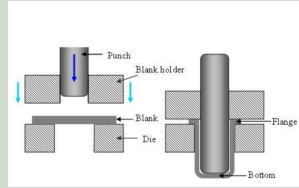
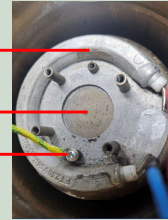


Diagram of how deep drawing occurs for a sheet of metal to gain the form of the milk frother shells.

Heating coil encase in a ceramic base.

Underside of the inner shell.

GND connection screwed to ceramic base.



3. Heater Plate Assembly

A ceramic heater plate is thermally bonded directly to the underside of the inner shell. This plate uses a ceramic substrate with a heating coil embedded within the plate itself.

The resistive coil is cast or printed into the ceramic plate which is directly connected to the inner shell, allowing direct transmission of heat.

The earth wire is secured to the heater plate's grounding point using a screwed ring terminal. The remaining two wires, live and neutral, are soldered directly to the heating coil ends on the ceramic plate.

A high-end Strix sensor is connected to the base of the inner chamber to accurately determine temperature.

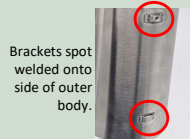
2. Component Refinement

Outer Shell

The outer shell has cut outs made for the display and cable routing into its body. As well as this, small metal brackets are positioned and spot welded onto the outer shell. Adding these fittings at this stage avoids heat damage to internal components later.

Inner Shell

The inner shell's surfaces are polished and flattened prior to having a ceramic reinforced coating applied to it. The coating is then heat-cured, hardening it and bonding it permanently to the stainless steel.



Brackets spot welded onto side of outer body.



Cut outs made into outer body.

This is a ceramic coating because the interior is matte, grippy and scratch-resistant. The characteristics of a cured ceramic layer.



The nesting process.

The rims of the top of the frother have soft edges with no laser marks, indicating this has been rolled rather than lasered into shape.



4. Body Assembly

Once the inner shell is fully prepared with the heater plate fitted, it is inserted into the outer shell.

The upper diameters align for rim joining, while the narrower lower section creates an annular air gap required for insulation and wiring clearance.

Rolled rim crimping is the assumed method used to secure the inner shell within the outer shell. This is evidenced by the rounded, tube-like profile as well as the lack of any weld seam or discoloration, ruling out the use of laser welding.

Furthermore, the spout is formed by folding the rim, which is only possible with crimped construction.

PCB Arrangement Analysis

Functional Zoning

The PCB was manufactured using functional zoning to streamline assembly and ensure electrical safety by separating the high-voltage power stage from the low-voltage control circuitry. This layout manages heat and reduces electrical noise, allowing both sections to coexist safely on a single board.

Large Components Come On Top

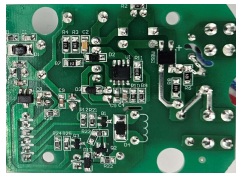
The top side carries **Through Hole power components** including large capacitors, transformers, relays and mains connections. These handle high voltage, generate heat and require physical stability and spacing.

Small Components Stay Below

On the underside of the PCB holds the **Surface Mount Technology (SMT)** control components such as resistors, transistors, diodes and small capacitors. These form the low-voltage logic of the system. The short track lengths and size of these parts allow them to be packed efficiently on the reverse side of the PCB.



Topside, with transformers, large capacitors and AC connections visible.

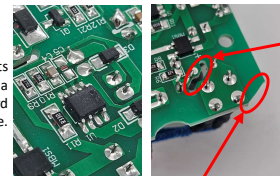


Underside, with transistors, resistors, diodes and small capacitors fitted.

Premium PCB Manufacture

The PCB shows evidence of high precision automated soldering and clear separation between high and low voltage areas.

Components soldered to a high standard by a machine.



Cutouts are added to the PCB during manufacture to separate high and low voltage areas by increasing creepage distance.

However, imperfections can be seen, such as poorly cut edges of the PCB board.

Improvements

Softer Edges For Safer Manufacture

During reassembly, the sharp edges of the metal base made it difficult to position the internal components correctly. These edges are a direct result of the deep-drawing process, as they were not rolled or finished after forming.

Introducing a slight rolled edge would reduce sharpness, improve handling during assembly, and lower the risk of injury for workers.



Very sharp edges can cause injury to workers during assembly.

Rolling these edges makes them less sharp and reduces the chance of injury at a low cost.



Use of Manual Production

Evidence of Manual Assembly

While components such as PCBs, motors, and internal frames clearly show machine production through precise soldering and consistent finishes, other elements display small imperfections indicating that certain assembly steps were likely performed by hand. As well as this, it is evident that individual components have been designed to be assembled manually, indicating this product is likely produced on a low to medium scale.

Misaligned Stickers



The sticker is visibly crooked, suggesting it was applied by hand rather than by a precision robot. Automated systems typically ensure consistent alignment.

Uneven metal grips for the handle



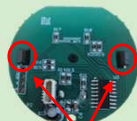
The misalignment of this grip indicates that this part was manually fitted to the body without the precision an automated system may have.

Plug-in connections between PCBs



These friction lock connections indicate manual assembly as they require precise alignment and are designed to be pushed in by hand.

PCB snap-fit clips



Snap-fit clips require the tabs to be manually squeezed and pushed through the PCB, so their use indicates this was assembled manually.

Automation Not Worth The Investment

Research into Morpilot's product range shows that their business model is built on the production of many small consumer products with short lifecycles and relatively low to medium production volumes.

As a result of frequent part and design changes, investing in dedicated automated assembly lines would be inefficient as the frequent process adjustments or retooling would drive up operational costs significantly.

Manual labour offers the flexibility needed to assemble varied products without expensive retooling or increased downtime to enact changes.

Implementing Additive Manufacturing

The internal frame and many other components are made of ABS, but it has several drawbacks:

- Costly tooling.
- Limited heat resistance.
- Poor environmental performance.

By using additive manufacturing with a suitable material such as Glass-Reinforced Nylon rather than injection moulding, Morpilot has the ability to support its strategy more effectively. Despite the higher upfront costs, additive manufacturing enables rapid design changes and suits their fast-moving, frequently updated product range without the constant need to replace old tools.

The material also provides higher stiffness, better heat resistance and a more reliable long-term performance than ABS.